LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Unit Test: Force and Pressure

Chapter / Topic: Force and Pressure (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 45 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks.
3. Write the unit with every numerical answer.

SECTION A

6 marks

Choose the correct option.

Q1. The SI unit of pressure is the: **[1]**

(a) newton (b) joule
(c) watt (d) pascal

Q2. Which of these is a non-contact force? **[1]**

(a) Friction (b) Muscular force
(c) Magnetic force (d) Tension in a rope

Q3. The pressure exerted by a liquid increases with: **[1]**

(a) depth (b) width of the container
(c) colour of the liquid (d) shape of the container

Q4. For the same force, pressure is greater when the area is: **[1]**

(a) larger (b) the same
(c) doubled (d) smaller

Q5. A force of 100 N acts on an area of 2 m^2. The pressure is: **[1]**

(a) 0.02 Pa (b) 50 Pa
(c) 102 Pa (d) 200 Pa

Q6. Which instrument measures atmospheric pressure? **[1]**

(a) Thermometer (b) Ammeter
(c) Barometer (d) Speedometer

SECTION B

8 marks

Q7. Give two examples showing that a force can change the shape of an object. **[2]**

Q8. Why do porters place a round piece of cloth on their heads when carrying heavy loads? **[2]**

Q9. Why do heavy trucks have broad tyres and many wheels? **[2]**

Q10. Which force makes a ball fall to the ground? Is it a contact or a non-contact force? **[2]**

SECTION C

6 marks

Q11. A box weighing 600 N rests on the floor on a base of area 1.5 m^2. Find the pressure on the floor. If the box is turned so that it rests on a face of area 0.5 m^2, what is the new pressure? **[3]**

Q12. Describe an activity to show that liquids exert equal pressure at the same depth. **[3]**

SECTION D

5 marks

Q13. Answer the following: **[5]**

(i) What is atmospheric pressure? Why are we not crushed by it? **[2]**

(ii) Give three examples from daily life where we use the idea of pressure, and explain one of them. **[3]**

~ All the best! ~